



AI-Powered Ransomware Detection System with Deep Learning and SOC-Based Risk Analysis

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Declaration

I declare that this dissertation, entitled

“AI-Powered Ransomware Detection System with Deep Learning and SOC-Based Risk Analysis,” is the result of my own independent work.

It has not been submitted previously, in whole or in part, for any academic degree or qualification at the University of Hertfordshire or any other institution. All sources of information used in the preparation of this work have been acknowledged appropriately and referenced in accordance with academic standards.

I confirm that this work complies fully with the University of Hertfordshire’s regulations on academic integrity and originality.

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1. Abstract

Ransomware continues to pose a significant cybersecurity risk since contemporary assaults might alter their configuration and conduct to evade traditional signature-based identification. This initiative created and assessed an AI-driven ransomware detection system that integrates machine learning, deep learning, and risk assessments from Security Operations Centers (SOCs). The primary objective was to examine the efficacy of differentiating ransomware from authentic software and how model forecasts can aid in prioritising security incidents.

The experimental dataset comprised 2,675 samples, consisting of 1,267 Goodware samples and 1,408 Ransomware samples. Subsequent to data preprocessing and exploratory analysis, three classification models were assessed: Random Forest, Deep Neural Network Model 1 (DNN1), and Deep Neural Network Model 2 (DNN2). The evaluation of model performance was conducted by confusion matrices, accuracy, precision, recall, ROC-AUC, Average Precision, and the examination of false positives and false negatives. Classification-threshold evaluation, ransomware-family retrieval, prediction-confidence assessment, and permutation feature significance were additionally conducted.

In the independent test set including 535 samples, Random Forest attained an accuracy of 98.32%, accurately identifying 280 out of 282 ransomware samples, resulting in merely 2 false negatives and 7 false positives. DNN1 attained an accuracy of 97.57% and a ransomware recall of 98.94%, with three false negatives, whilst DNN2 reached an accuracy of 97.57% and a recall of 97.87%. The ROC-AUC values for Random Forest, DNN1, and DNN2 were 0.9997, 0.9947, and 0.9952, respectively. Threshold research revealed a distinct compromise between minimising false alarms and augmenting overlooked ransomware incidents.

The findings indicate that deep learning offers exceptionally efficient ransomware detection, despite the Random Forest baseline demonstrating the most robust overall performance. Incorporating prediction certainty with SOC-focused risk classifications further illustrated how AI detection may aid in prioritising incidents. Nonetheless, reliance on datasets, variations at the family level, and significantly erroneous predictions suggest that the system ought to assist rather than supplant skilled SOC analysts.

2. Introduction

2.1 Problem Overview

Ransomware is a major cybersecurity threat to modern businesses. It is malicious software that encrypts, locks, steals, or restricts data and systems and demands payment. Ransomware assaults now include data theft, extortion, and security monitoring preventance, complicating the issue. In 2024, ENISA classified ransomware as one of the top cybersecurity dangers. At least 1,000 ransomware claims per quarter were reported on data-leak sites in part of 2024 (ENISA, 2024).

The effects of ransomware go beyond technological interruption. Successful attacks can disrupt company, customer service, and application access, causing financial and reputational damage. IBM found that the global average data breach cost USD 4.88 million in 2024, up 10% from the year before. After a compromise, 70% of companies reported severe or very significant operational impact (IBM, 2024). The average breach cost INR 195 million in India in 2024, demonstrating that cybersecurity incidents can have significant economic impacts across markets (IBM, 2024). These numbers show why ransomware detection must be improved for research and business.

Conventional anti-malware methods rely on signatures or patterns. While these approaches work for known threats, they may not work for updated, disguised, or impersonating legitimate activity ransomware. Defensive-evasion strategies like trusted services and legitimate system tools are being used more to hide malicious activities (ENISA, 2024). This requires detection approaches that look beyond signs to examine structural and behavioural traits.

Instead, Machine Learning (ML) and Deep Learning (DL) can detect statistical associations across numerous features and classify observations based on past data trends. This lets cybersecurity models identify legitimate software from malware based on structure or activity. AI raises issues including inaccurate predictions, model transparency, dataset quality, and overconfidence. Thus, the current project does not judge success solely by accuracy. False positives, false negatives, ransomware recall, ROC-AUC, Average Precision, classification thresholds, and prediction confidence are explored.

2.2 Current Issues and Research Motivation

A primary challenge in AI-driven cybersecurity research is that elevated model accuracy does not inherently indicate operational dependability. In the realm of ransomware detection, two categories of errors are especially significant. A false positive arises when authentic software is misidentified as ransomware, while a false negative transpires when ransomware is misclassified as legitimate software. Erroneous positives can elevate the burden on Security Operations Center (SOC) analysts, however erroneous negatives pose a greater threat as harmful actions may persist unexamined.

This matter is significant when evaluating the magnitude of contemporary security activities. The Verizon 2025 Data Breach Investigations Report examined over 22,000 security events and 12,195 verified breaches spanning 139 nations. In the Asia-Pacific area, ransomware accounted for 51% of breaches, indicating its continued significance in contemporary security defence strategies (Verizon, 2025).

A further significant concern is the presumption that deep learning would invariably surpass traditional machine-learning techniques. Deep neural networks are capable of representing intricate non-linear associations; nonetheless, they often necessitate more computer power, meticulous training, and substantial quantities of representative data. A more straightforward technique might yield comparable results or perhaps

superior performance when the dataset is organised and tabular. Consequently, this initiative evaluates two Deep Neural Network architectures against a Random Forest benchmark instead of presuming that a deep learning model will inherently yield superior results.

The quality of the dataset is a significant issue. Cybersecurity datasets could have redundant entries, highly correlated attributes, class disparities, or traits that predominantly indicate the dataset's creation process rather than authentic hostile activity. Such problems can yield implausible model efficacy. Consequently, the current study employs exploratory analysis, class distribution analysis, feature correlation analysis, and permutation importance in conjunction with predictive modelling. The objective is to comprehend the reasons behind the models' efficacy and to discern potential constraints in generalisation.

An further impetus for study pertains to the pragmatic use of AI forecasts. A binary classification like Goodware or Ransomware does not adequately address the operational inquiry confronted by a SOC analyst: Which alert warrants immediate investigation? The NIST Cybersecurity Framework 2.0 elucidates that proficient cybersecurity encompasses not merely detection but also synchronised reaction and recovery efforts (NIST, 2024). The existing project thus enhances model classification by integrating a SOC-based risk component that translates forecasts into Low, Medium, High, and Critical risk classifications.

2.3 Project Details and Technical Focus

The initiative is named AI-Enhanced Ransomware Detection Framework utilising Deep Learning and SOC-Centric Risk Assessment. It examines the application of structural software characteristics to distinguish Goodware from Ransomware and integrates prediction outcomes with SOC-focused risk prioritisation.

The finalised solution utilises an experimental dataset comprising 2,675 samples, which includes 1,267 Goodware samples and 1,408 Ransomware samples. This signifies roughly 47.36%. Goodware constitutes 52.64% of the dataset, resulting in a rather equitable binary classification challenge.

Three primary classification models were assessed. The initial model employed was a Random Forest, serving as a standard machine-learning benchmark. Two configurations of Deep Neural Networks, designated as DNN Model 1 and DNN Model 2, were subsequently created and evaluated against the baseline.

The ultimate independent test dataset comprised 535 instances, consisting of 253 Goodware and 282 Ransomware samples. The Random Forest algorithm accurately identified 246 Goodware and 280 Ransomware samples, resulting in merely 7 false positives and 2 false negatives. DNN Model 1 yielded 243 true negatives, 10 false positives, 3 false negatives, and 279 true positives, whereas DNN Model 2 yielded 246 true negatives, 7 false positives, 6 false negatives, and 276 true positives.

The models demonstrated exceptionally high discrimination efficacy. The Random Forest model attained a ROC-AUC of 0.9997, while DNN Model 1 reached 0.9947 and DNN Model 2 scored 0.9952. The Average Precision values were 0.9997, 0.9937, and 0.9950, in that order. These results underscore the significance of model comparison inside the project, as the most robust model was not the most intricate.

Alongside these fundamental assessments, the initiative investigated classification-threshold dynamics, ransomware-family recall, the confidence associated with accurate and erroneous predictions, structural-feature correlations, and the significance of permutation features. A SOC-focused risk framework was subsequently utilised on the ultimate predictions to categorise test observations into Low, Medium, High, and Critical risk tiers.

2.4 Aim and Objectives

The main aim of the project is:

To develop and critically evaluate an AI-based ransomware detection framework that uses machine learning and deep learning to distinguish ransomware from legitimate software and integrates prediction results into an SOC-oriented risk-prioritisation process.

The project objectives are:

1. To provide a thorough evaluation of contemporary studies on ransomware, behavioural malware identification, machine learning, deep learning, and security operations centers.
2. To compile and examine a suitable ransomware dataset that encompasses the structural attributes of Goodware and Ransomware specimens.
3. Conduct data preparation and Exploratory Data Analysis to comprehend class distribution, feature correlations, and potential ransomware indicators.
4. To establish a Random Forest model as a standard machine-learning reference.
5. To formulate and assess two configurations of Deep Neural Networks for the purpose of ransomware detection.
6. To evaluate model efficacy utilising Accuracy, Precision, Recall, F1-Score, ROC-AUC, Average Precision, and confusion matrices.
7. To examine the impact of altering the decision threshold on ransomware recall, as well as the rates of false positives and false negatives.
8. To investigate if the efficacy of the approach is uniform across various ransomware families.
9. To discern significant structural characteristics through correlation and permutation significance.
10. To include forecasts and confidence data into SOC-oriented risk classifications.
11. To assess the dependability, constraints, and practical significance of the ultimate prototype.

2.5 Research Question and Novelty

The main research question is:

How effectively can deep learning models detect ransomware-related malicious behaviour, and how can model predictions be integrated with a risk-scoring framework to support SOC incident prioritisation?

This inquiry is bolstered by multiple subsidiary queries. The study examines the structural traits that differentiate Goodware from Ransomware, the comparative efficacy of DNN models versus Random Forest, the impact of classification thresholds on security inaccuracies, the consistency of ransomware-family performance, and the role of prediction confidence in SOC prioritisation.

The originality of the study mostly lies in the integrated assessment methodology, rather than in introducing an entirely new deep learning framework. Numerous cybersecurity studies emphasise attaining a superior categorisation score. This initiative, in contrast, assesses the models from multiple operational viewpoints. It contrasts traditional and deep learning techniques, examines the trade-offs between false alerts and missed attacks, evaluates performance at the ransomware family level, analyses confidence in inaccurate predictions, explores significant features, and correlates model results with SOC risk classifications.

This fosters a more pragmatic viewpoint as the initiative enquires not only about the detectability of ransomware but also whether the forecasts yield valuable insights for operational cybersecurity choices.

2.6 Feasibility, Commercial Context and Risk

The initiative was conceived as a research prototype to ensure it stayed feasible within the existing academic schedule and computational capabilities. Python-centric instruments facilitated data processing, modelling, and assessment, while widely accessible machine learning and deep learning libraries aided in model creation. This diminished the necessity for costly commercial security systems and facilitated reproducible experimentation.

From a business standpoint, enhanced ransomware detection may significantly diminish operational interruptions, investigative expenses, and the broader financial repercussions of successful assaults. The significance of efficient incident response in economic terms is underscored by IBM's 2024 research, which indicates that businesses that engage law enforcement during ransomware events incur breach costs roughly USD 1 million less on average, not including ransom payments (IBM, 2024). While the existing prototype cannot assert actual cost benefits, the data illustrates the business significance of expedited detection and prioritised treatment.

Numerous hazards also impact the project. A model might excel on the chosen dataset yet struggle to adapt to novel malware. Certain structural characteristics might pertain exclusively to the dataset instead than being universally applicable malware indications. Erroneous positives might generate superfluous workload for the SOC, whereas erroneous negatives signify undetected ransomware. The reliability of prediction confidence can be deceptive, since experimental findings indicate that certain erroneous classifications were assigned elevated confidence levels.

Consideration must also be given to ethical and professional hazards. NIST's AI Risk Management Framework underscores the necessity of assessing AI systems concerning validity, dependability, security, transparency, and the risks associated with their deployment environment (NIST, 2023). As a result, the established system is regarded as a prototype that assists analysts rather than serving as an independent substitute for expert security decision-making.

2.7 Scope and Limitations

The project's focus is confined to binary classification distinguishing Goodware from Ransomware utilising the chosen study dataset. Ransomware-family IDs are employed for enhanced robustness evaluation, however the system refrains from doing comprehensive malware-family attribution.

The initiative does not utilise active ransomware, gather information from operational endpoints, or autonomously segregate organisational systems. These limitations enhance practicality and mitigate ethical and security concerns. Nevertheless, they also restrict the degree to which experimental findings may be extrapolated to actual SOC settings.

Despite the exceptionally robust results, featuring a Random Forest ROC-AUC of 0.9997, such elevated performance warrants cautious interpretation. External validation using independent datasets, newly acquired ransomware samples, and authentic organisational settings is necessary before the methodology can be deemed appropriate for practical use.

2.8 Report Structure

The subsequent sections of the report elaborate on the subjects presented in this chapter with increased detail. Section 3 presents the Literature Review and rigorously analyses ransomware detection, machine learning, deep learning, feature analysis, model assessment, and SOC-oriented security studies. Section 4 delineates the Methodology, encompassing dataset preparation, preprocessing, model construction, experimental design, threshold evaluation, and risk assessment. Section 5 delineates the Quality and Results, encompassing confusion matrices, ROC and precision-recall metrics, ransomware-family evaluations, confidence assessments, feature significance, and SOC risk outcomes. Section 6 rigorously assesses the results in relation to the research aims and current literature, examines project management, constraints, economic significance, and prospective advancements, and presents the concluding remarks. Section 7 has the Harvard-style citations, but Section 8 offers supplementary appendices that encompass technical proof, code snippets, and further experimental results.

This introduction delineates ransomware detection as a technical categorisation issue and a practical cybersecurity obstacle. The initiative assesses not just the efficacy of AI in identifying ransomware but also the interpretability and prioritisation of its predictions for SOC decision-making.

3. Literature Review

3.1 Introduction to Ransomware Detection Research

Ransomware constitutes a type of malicious software intended to restrict access to data or systems, typically via encryption, while soliciting payment from the affected party. Contemporary ransomware has evolved past just file encryption and may encompass data exfiltration, dual extortion, credential exploitation, and efforts to undermine security measures. This advancement has rendered ransomware detection a critical research issue, as identifying it post-encryption may be insufficient to avert significant harm. Cen et al. (2024) contend that ransomware ought to be recognised prior to any harmful actions occurring, thus establishing early detection as a pivotal necessity for contemporary ransomware protection.

Consequently, research has transitioned from exclusively signature-based identification to behavioural analysis, machine learning (ML), and deep learning (DL). Urooj et al. (2022) discovered that machine learning and deep learning have been progressively utilised for ransomware detection through the use of dynamically gathered attributes, such as API calls, system calls, file activities, and various runtime properties. Their assessment also highlights dataset integrity, computational intricacy, and the ever-evolving nature of ransomware groups as significant outstanding issues.

The ongoing effort is situated inside this research domain, although it specifically emphasises the correlation between classification efficacy and operational security worth. Instead of solely assessing an AI model's ability to distinguish between Goodware and Ransomware, the research contrasts Random Forest with two Deep Neural Networks (DNNs), explores classification thresholds, analyses ransomware-family recall and prediction confidence, scrutinises significant features, and translates predictions into SOC-focused risk categories. The literature is consequently examined with these project aims in consideration.

3.2 Traditional and Behaviour-Based Ransomware Detection

Conventional malware detection typically depends on established signatures or protocols. Signature-based detection analyses the attributes of an executable or action against established patterns linked to harmful software. This method is computationally efficient and retains its significance when the malware is already identified. Nonetheless, its reliance on previously recognised patterns renders it vulnerable to novel or altered malware.

Contemporary ransomware can employ polymorphism, metamorphism, and obfuscation to alter its apparent traits while maintaining its harmful capabilities. A recent analysis of deep-learning ransomware detection highlights these methods as significant factors contributing to the growing challenge traditional anti-malware systems face in identifying complicated ransomware variants (Kaur et al., 2024). Consequently, identifying ransomware just through the correlation of established signatures may yield insufficient security.

Behavioural and structural analysis seeks to overcome this constraint by investigating the contents or functions of software instead of merely assessing its conformity to a recognised malware signature. Dynamic analysis can monitor API invocations, system calls, file system operations, registry modifications, and various runtime activities. Urooj et al. (2022) conducted a study of ransomware detection research employing dynamic analysis and discovered a significant use of these characteristics with both machine-learning and deep-learning classifiers.

Nevertheless, dynamic analysis possesses some constraints. Malware can recognise its execution within a sandbox environment and postpone its harmful actions. Urooj et al. (2022) cite prior studies in which samples were executed for merely ten seconds, allowing concealed ransomware to evade detection of its destructive activities throughout the monitoring phase. Static and structural analysis circumvents certain runtime

expenses, however it can be influenced by obfuscation. These results indicate that no individual feature source should be presumed to be universally dependable.

This matter pertains to the ongoing project as structural attributes are employed to differentiate Goodware from Ransomware. Robust correlations between some structural factors and the ransomware target can enhance classification; however, very strong associations should be scrutinised to ascertain if they reflect authentic malware traits or are merely aberrations of the dataset.

3.3 Machine Learning for Ransomware Detection

Machine learning offers a data-centric substitute for rigid detection protocols. Rather than manually defining each ransomware pattern, machine learning models discern correlations between input characteristics and designated results. Common ransomware classifiers include Logistic Regression, Decision Trees, Random Forest (RF), Support Vector Machines (SVM), k-Nearest Neighbours and ensemble approaches.

A recent examination of machine learning-driven ransomware detection indicates that experimental research has yielded detection accuracies between roughly 81.38% and 96.28%, though direct comparisons among studies are challenging due to variations in datasets, feature types, preprocessing, and validation methods (Almomani et al., 2023). This variation illustrates that the accuracy presented by one study cannot be directly compared to another without taking the experimental design into account.

Random Forest holds significant relevance to the present inquiry. Random Forest amalgamates several decision trees and consolidates their results. This can represent non-linear associations while still being efficient with organised tabular data. It necessitates less architectural design compared to a neural network and can yield valuable feature-importance insights.

The significance of maintaining a traditional ML baseline is occasionally undervalued when deep learning dominates the research emphasis. A more intricate model should only be deemed superior provided empirical evidence substantiates the added complexity. This notion had significant relevance in the current project: the ultimate Random Forest attained a ROC-AUC of 0.9997, in contrast to 0.9947 for DNN Model 1 and 0.9952 for DNN Model 2. Consequently, the initiative contests any basic presumption that deep learning is inherently superior to traditional machine learning.

Nonetheless, there are constraints to ransomware detection based on machine learning. The efficacy is heavily influenced by the training distribution and the engineering of features. Models might acquire associations unique to a particular dataset and demonstrate diminished efficacy when confronted with novel ransomware variants or altered behaviours. Urooj et al. (2022) characterise this as an issue of population drift due to the ongoing evolution of ransomware variants. Thus, autonomous evaluation and familial-level examination are significant enhancements to comprehensive classification metrics.

3.4 Deep Learning for Ransomware Detection

Deep learning uses multi-layered neural networks to learn complicated non-linear correlations. Numerous ransomware studies have examined DNNs, CNNs, RNNs, LSTMs, autoencoders, and hybrid architectures.

Deep learning can model complex feature associations without manual decision criteria, making it appealing. A comprehensive 2024 evaluation discovered ransomware deep-learning studies with 91%–99.8% accuracy. CNNs, RNNs, LSTMs, and Deep Belief Networks performed well (Kaur et al., 2024).

High values need critical interpretation. Ransomware families, class distributions, sample sizes, and validation methods vary across study. A model with 99% accuracy on a controlled dataset may not perform well against new malware in a live network. DL-based ransomware detection still struggles with unlabelled datasets, computational cost, and black-box interpretability, according to the same review.

This greatly supports the project's design. Instead of reporting one successful DNN network, two are examined. They also compare to Random Forest. DNN Model 1 detected 279 of 282 ransomware samples on the final test set, yielding 98.94% recall, whereas Model 2 spotted 276. These results reveal great DL performance, commensurate with recent literature's high detection rates, but the stronger RF result shows why comparative evaluation is important.

Attention is needed for behaviour training. Overfitting occurs when neural networks learn non-generalising training-specific patterns. The current DNN Model 2 experiment showed that training and validation loss initially reduced jointly, but validation loss eventually increased while training loss declined. This shows why learning curves and independent tests are more informative than training accuracy.

3.5 Dataset Quality, Features and Generalisation

The efficacy of a ransomware detection system is intricately tied to the calibre of the data utilised in its development. An optimal dataset must encompass representative authentic and nefarious instances, significant attributes, and adequate diversity within ransomware categories. It must also reduce redundant observations and prevent information leakage between the training and testing datasets.

Contemporary research consistently highlights the lack of appropriate public benchmark datasets as a major obstacle. Urooj et al. (2022) observe the absence of a unified benchmark dataset that facilitates reliable comparisons among ransomware detection methodologies. Kaur et al. (2024) similarly recognise that the scarcity of pertinent tagged examples remains a persistent constraint in deep-learning ransomware studies.

This issue influences the understanding of elevated classification outcomes. If samples that are closely linked are present in both training and testing datasets, the model might proficiently identify similarities instead of acquiring patterns that apply to unfamiliar malware. Attributes may inadvertently encapsulate class information. Consequently, very robust predicted correlations necessitate scrutiny instead of being instantly regarded as proof of a superior classifier.

The dataset employed in this experiment comprises 2,675 observations: 1,267 Goodware samples and 1,408 Ransomware samples. The disparity among the classes is rather minor, diminishing the probability that accuracy is only influenced by a prevailing majority class. Exploratory study, however, revealed significant correlations between ransomware and structural characteristics including `_initterm`, `_xcpfilter`, and `_exit`. These associations yield valuable prediction insights, although they also warrant meticulous scrutiny for possible dataset-specific influences.

The initiative therefore integrates correlation analysis with permutation significance. These techniques address various enquiries. Correlation quantifies the statistical association between a certain variable and the target, whereas permutation importance assesses the alteration in model efficacy when a feature's data is perturbed. In the DNN analysis, the permutation of `.rsrc` resulted in the most significant reduction in F1-score, succeeded by

SECURITY_Size and getcommandlinea. These findings are regarded as predicted correlations, rather than evidence that these characteristics actively induce ransomware activity.

3.6 Ransomware-Family Robustness, Confidence and Explainability

The overall efficacy may obscure inadequate identification of specific ransomware variants. This is significant as a model could excel when the test distribution closely mirrors the training data but struggle with a less prevalent or structurally dissimilar category. The evolving ransomware demographic highlighted by Urooj et al. (2022) underscores the necessity for family-level and prospective distribution assessments.

The ongoing experiment thus evaluates recollection concerning ransomware-family IDs. The majority of evaluated families attained a recall of 1.00, but family IDs 28, 23, and 15 yielded recall rates of roughly 0.75, 0.83, and 0.89, respectively. This outcome is significant as the overall performance of the DNN would obscure these deficiencies. It additionally aligns with the broader scholarly discourse around the generalisation to evolving ransomware variants.

The confidence of the model poses an additional challenge. A classifier may yield a probability nearing 1.0 without being accurate. In the present studies, accurate predictions were clustered near peak confidence, while erroneous predictions varied roughly between 0.71 and 1.00, with a median around 0.96. Consequently, elevated confidence ought not to be regarded as absolute certainty.

This is strongly related to the interpretability issue highlighted in contemporary deep-learning research. Kaur et al. (2024) highlight the opaque characteristics of deep learning models as an ongoing issue in ransomware identification. Feature-importance techniques can partially resolve this issue by indicating which variables affect predictions; nevertheless, they cannot establish causal relationships. Consequently, human interpretation is still essential.

3.8 SOC Risk Prioritisation and Research Gap

The majority of ransomware detection studies focus on the accuracy of a model in recognising harmful behaviour. Operational cybersecurity necessitates an extra measure. SOC analysts may encounter numerous warnings and must determine which incidents necessitate prompt examination. A classifier that produces probabilities without a discernible priority ranking consequently offers merely a fraction of the necessary information for event response.

Contemporary ransomware studies are progressively acknowledging the importance of including detection into broader cybersecurity structures. Kaur et al. (2024) see the incorporation of incident-response frameworks and threat-intelligence systems as a crucial avenue for enhancing the operational efficacy of ransomware detection. Urooj et al. (2022) concurrently underscore the necessity for additional research in computing efficiency, evolving ransomware demographics, and practical detection limitations.

The reviewed literature reveals four pertinent gaps that are especially significant to this endeavour. Initially, the significance of superior classification performance is often highlighted, although there is a scarcity of research investigating the operational implications of categorisation thresholds. Secondly, collective performance may obscure the inadequate identification of certain ransomware variants. Third, the

interpretation of model confidence and feature significance necessitates meticulous consideration, as predictions that exhibit high confidence or seem explicable may nevertheless be erroneous. Firstly, there exists potential for a more integrated relationship between AI-driven ransomware identification and SOC-focused risk assessment.

4. Methodology

4.1 Research and Project Management Approach

This initiative employed a quantitative experimental approach as the primary research inquiry necessitated a measurable evaluation of ransomware detection methods and their appropriateness for Security Operations Center (SOC) risk prioritisation. The inquiry included data analysis, machine learning, deep learning, model validation, explainability and risk grading within one technical artefact. This methodology was suitable as contemporary ransomware studies indicate that effective AI-driven detection relies on multiple interrelated phases, such as data preparation, feature engineering, model training, and performance assessment (Ferdous et al., 2024).

A blended project management methodology was employed to oversee the operational tasks. The entire project was structured around designated milestones, whereas execution adhered to a cyclical methodology. The primary stages encompassed literature review, dataset curation, Exploratory Data Analysis (EDA), baseline model creation, deep-learning experimentation, model evaluation, threshold assessment, rigorous validation, SOC risk evaluation, and conclusive reporting. This framework aligned with the project blueprint established during the IPR, which segmented implementation into phases of data preparation, model building, risk assessment, testing, and reporting.

The iterative part was vital since machine-learning studies rarely follow a perfectly linear approach. EDA uncovered significant feature interrelations, model evaluations demonstrated varying false-positive and false-negative tendencies, and subsequent tests necessitated threshold and ransomware family assessments. Consequently, the approach facilitated the re-examination of prior phases when empirical results necessitated additional scrutiny.

4.2 System and Experimental Design

This design formulated the initial project proposal, which outlined a preprocessing pipeline, deep-learning detection, model assessment, risk rating, and SOC-focused alert prioritisation. The ultimate execution was enhanced in accordance with the attributes of the accessible data. The study specifically focused on Random Forest and feed-forward Deep Neural Networks instead of utilising LSTM due to its complexity as a deep-learning method. LSTM is largely intended for sequential data, while the ultimate project dataset is predominantly composed of organised software attributes. Selecting models according to the data structure enhanced methodological justification.

The experimental framework additionally incorporated a traditional machine-learning benchmark. Random Forest was chosen due to the robust performance of ensemble tree techniques on structured tabular data, since they may capture non-linear relationships without the need for complex scaling and architectural considerations typical of neural networks. The DNN models were then assessed against this baseline. This comparison was crucial as current evaluations caution that intricate deep-learning techniques ought to be assessed alongside basic options instead of being presumed superior (Ferdous et al., 2024).

4.3 Dataset Preparation and Data Characteristics

The ultimate refined dataset comprised 2,675 observations, featuring 1,267 Goodware instances and 1,408 Ransomware instances. Goodware constituted roughly 47.36% of the observations, whereas Ransomware accounted for 52.64%. The minor disparity across the classes diminished the likelihood that the overall accuracy would be overshadowed by a single predominant category.

Prior to modelling, the data was scrutinised for organization, feature categories, absent information, redundancies, and class distribution. This phase was crucial as unreliable cybersecurity datasets can yield deceptively robust model outcomes. Recent evaluations of AI-driven ransomware detection highlight preprocessing, feature engineering, and dataset quality as critical elements influencing model dependability (Ferdous et al., 2024).

Exploratory Data Analysis was subsequently conducted to examine the correlation between existing structural characteristics and the ransomware target. Numerous attributes, such as `_initterm`, `_xcpfilter`, `_exit`, `exit`, `msvcrt.dll`, and `.rsrc`, had significant correlations with the objective. These results bolstered further modelling however were not regarded as causal proof. Robust feature-target correlations were deemed essential since cybersecurity models might occasionally acquire dataset-specific artefacts instead of behaviours that generalise to unfamiliar malware.

The ultimate test dataset comprised 535 instances, including 253 Goodware and 282 Ransomware samples. This concealed test set was preserved for the ultimate assessment to ensure that performance metrics were not only based on the data utilised in model construction.

4.4 Data Preprocessing and Feature Preparation

The pretreatment methodology was established to generate a uniform dataset suitable for modelling, while minimising unnecessary biases. The preliminary examination was succeeded by data sanitisation, target formulation, and numerical attribute processing. Duplicate and erroneous records were examined, and inappropriate identifier data was omitted where required. Numerical attributes were formatted to align with the chosen algorithms.

The significance of scaling is especially pertinent to neural networks, as features with vastly differing numerical ranges can adversely affect gradient-based optimisation. Random Forest has reduced sensitivity to feature scaling since its decisions rely on feature splits, although uniform preprocessing facilitates systematic model comparison.

The division of training, validation, and testing was employed to differentiate model development from the ultimate evaluation. The training segment enabled the algorithms to acquire model parameters. Validation observations were employed to oversee neural network training and evaluate intermediate outcomes, whereas the independent test set facilitated the ultimate assessment. This method mitigated the likelihood of presenting excessively favourable outcomes due to the continuous assessment of performance on identical data.

4.5 Baseline Random Forest Model

The Random Forest algorithm was initially employed as the foundational classifier. The aim of this experiment was to create a solid benchmark prior to assessing the increased intricacy of deep learning. A multitude of

decision trees were amalgamated to ensure that the ultimate prediction represented an ensemble rather than an individual tree.

The baseline additionally bolstered methodological validity as the research inquiry focuses on the efficacy of AI-driven detection rather than establishing that deep learning is invariably superior. This differentiation proved significant during the evaluation. On the autonomous test dataset, Random Forest accurately categorised 526 out of 535 instances, yielding an accuracy rate of roughly 98.32%. It yielded 246 true negatives, 7 false positives, 2 false negatives, and 280 true positives. The ransomware recall was thus roughly 99.29%, indicating that a traditional ensemble model established a robust standard.

4.6 Deep Neural Network Development

Two distinct DNN configurations were created to examine if alterations in neural network architecture influenced generalisation. Feed-forward DNNs were suitable as they may represent intricate non-linear associations among organised input variables without necessitating temporal order. Recent studies recognise DNNs as beneficial for cybersecurity classification, while also highlighting the importance of meticulous architecture selection, regularisation, and validation (Kaur et al., 2024).

DNN Model 1 underwent training with continuous observation of training and validation metrics across the epochs. The precision rapidly improved in the initial training phase and then stabilised in the high-98% to 99% spectrum. The validation performance was nearly equivalent to the training performance, indicating effective generalisation during the majority of the training period.

DNN Model 2 was created as an additional experiment instead of simply reiterating the initial model. The precision improved significantly; nevertheless, the validation loss curve started to rise marginally in the later epochs, even while the training loss persisted in its decline. This conduct indicated little overfitting in the later stages. Thus, it was essential to observe both accuracy and loss, as ultimate accuracy alone would not have disclosed this tendency.

In the independent test set, DNN Model 1 yielded 243 true negatives, 10 false positives, 3 false negatives, and 279 real positives, resulting in an accuracy of roughly 97.57% and a ransomware recall of 98.94%. DNN Model 2 yielded 246 true negatives, 7 false positives, 6 false negatives, and 276 true positives, resulting in an accuracy of roughly 97.57% and a ransomware recall of 97.87%.

4.7 Evaluation Metrics and Test Strategy

The assessment was structured to evaluate both statistical efficacy and real-world cybersecurity practices. Solely relying on accuracy was deemed inadequate due to the markedly different repercussions of false positives and false negatives. Contemporary ransomware studies advocate for a multi-metric assessment instead of reliance on a singular prominent outcome (Kaur et al., 2024).

The primary measures included Accuracy, Precision, Recall, F1-Score, ROC-AUC, Average Precision, and confusion matrices. The significance of recall lay in its ability to quantify the effective identification of ransomware incidents. False negatives denoted ransomware specimens that the system erroneously categorised as Goodware, whilst false positives indicated authentic samples that mistakenly triggered malicious alerts.

ROC analysis assessed differentiation across decision thresholds. The Random Forest attained a concluding ROC-AUC of 0.9997, whereas DNN Model 1 reached 0.9947 and DNN Model 2 scored 0.9952. The precision-recall assessment yielded Average Precision figures of 0.9997, 0.9937, and 0.9950, correspondingly. These evaluations revealed that all models exhibited robust performance; however, increased model complexity did not yield superior overall discrimination.

4.8 Threshold and Advanced Robustness Testing

A distinct threshold experiment was conducted since the standard probability threshold may not yield the optimal cybersecurity operational point. DNN probabilities were assessed across various thresholds ranging from around 0.20 to 0.80. Reduced thresholds preserved exceptionally high ransomware detection rates but resulted in an increase in false positives. As the threshold was raised, the incidence of false positives diminished, however false negatives escalated. At elevated tested levels, the number of false positives decreased to roughly three, but the instances of missed ransomware rose to about five.

This compromise illustrated that threshold determination ought to be contingent upon operational priorities. An entity that faces significant expenses from a failed ransomware assault may opt for a diminished threshold, despite SOC analysts receiving more notifications.

Validation at the family level was also performed, as collective measurements may obscure deficiencies. The majority of ransomware families attained a recall rate of 1.00, whereas family IDs 28, 23, and 15 reached roughly 0.75, 0.83, and 0.89, respectively. These findings demonstrated that superior overall performance did not ensure uniform reliability among all malware families.

The trust in predictions was additionally examined by contrasting accurate and inaccurate classifications. Accurate forecasts were predominantly clustered around a confidence level of 1.0, but erroneous predictions varied from roughly 0.71 to almost 1.00, with a median near 0.96. As a result, confidence was regarded as valuable supplementary data rather than evidence of the accuracy of a prediction.

4.9 Feature Interpretation and SOC Risk Analysis

Permutation significance was utilised to enhance comprehension of the DNN. The disruption of each chosen feature was analysed, revealing the drop in F1-score. .rsrc exhibited the most significant decline in performance, succeeded by SECURITY_Size and getcommandlinea. Additional significant factors encompassed .data, version.dll, SizeOfStackReserve, regclosekey, Machine, and SizeOfOptionalHeader.

Permutation importance was chosen as it evaluates the correlation between characteristics and their genuine predictive efficacy. Nonetheless, feature significance was not construed as causative. A factor that aids in classification does not inherently serve as a direct cause or universal sign of ransomware.

The ultimate model results were integrated into a risk component focused on SOC. Forecasts and confidence data were classified into Low, Medium, High, and Critical risk categories. The majority of observations were found in the Low and Critical categories, with a limited number of Medium and High instances. This indicated that the existing scoring system frequently resulted in distinctly divergent risk assessments. NIST CSF 2.0 underscores the necessity for a systematic approach to identifying, assessing, prioritising, and communicating

cybersecurity risks (NIST, 2024). The risk layer therefore established a tangible link between model forecasts and incident priority.

4.10 Tools, Validation and Reliability

Python was selected as the primary programming language due to its comprehensive ecosystem for data processing, machine learning, deep learning, and visualisation. Pandas and NumPy facilitated data manipulation, Scikit-learn enabled preprocessing, Random Forest provided modelling and assessment, and TensorFlow/Keras assisted with neural network construction. Matplotlib was employed for performance analysis and exploratory data visualisation. Notebook-centric development facilitated progressive experimentation and record-keeping.

Reliability was enhanced using many validation methods instead of a single assessment. These encompassed validation-set oversight, a separate conclusive test set, confusion-matrix assessment, ROC and precision-recall appraisal, two DNN architectures, juxtaposition with Random Forest, threshold sensitivity examination, family-level recall, confidence evaluation, and permutation significance. This triangulation diminished reliance on a solitary performance metric.

Nonetheless, the outcomes are still confirmed internally within the chosen dataset. Extremely elevated figures, like the Random Forest ROC-AUC of 0.9997, necessitate careful analysis. Conducting external testing on a disparate ransomware dataset would yield more compelling proof that the identified associations extend beyond the existing data.

4.11 Ethical, Legal, Professional and Practical Considerations

The initiative utilised secondary cybersecurity information and did not necessitate the direct gathering of personal data from individuals. No ransomware was utilised on external systems, and the studies were confined to defensive evaluation. This diminished ethical and operational risk. The NIST AI Risk Management Framework asserts that AI systems must be created and assessed with a focus on validity, dependability, transparency, and risk management (NIST, 2023). Consequently, the prototype is seen as a tool for decision support instead of a self-sufficient substitute for SOC analysts.

Legal factors encompass the proper utilisation of datasets, software licenses, intellectual property rights, and forthcoming data protection obligations. An operational SOC deployment may handle IP addresses, employee actions, or device data, necessitating enhanced privacy safeguards.

The primary practical constraint is external validity. Outcomes derived from a regulated dataset may not accurately reflect evolving ransomware patterns, idea drift, or hostile interference. Errors with high confidence and diminished recall for certain families underscore this constraint. The allocation of computational resources, development duration, and project parameters necessitated the prioritising of the most robust trials rather than executing all initially contemplated models.

The approach integrated regulated experimentation with comparative and operational assessments. The utilisation of independent test data, diverse evaluation metrics, threshold analysis, family-level validation,

confidence assessment, and SOC risk analysis offered a more robust foundation for addressing the research topic than only presenting classification accuracy.

5. Quality and Results

5.1 Overview of Experimental Quality

This chapter delineates the primary experimental findings of the ransomware detection system and assesses their quality, dependability, and practical significance. The outcomes are associated with the project's aims of differentiating Goodware from Ransomware, contrasting traditional machine learning with deep learning, examining false positives and overlooked attacks, pinpointing significant features, and facilitating SOC-focused risk prioritisation. The conclusive dataset comprised 2,675 samples, consisting of 1,267 Goodware samples and 1,408 Ransomware samples. This resulted in a very equitable class distribution of 47.36% Goodware and 52.64% Ransomware, diminishing the probability that a high overall accuracy score would stem only from forecasting the predominant class.

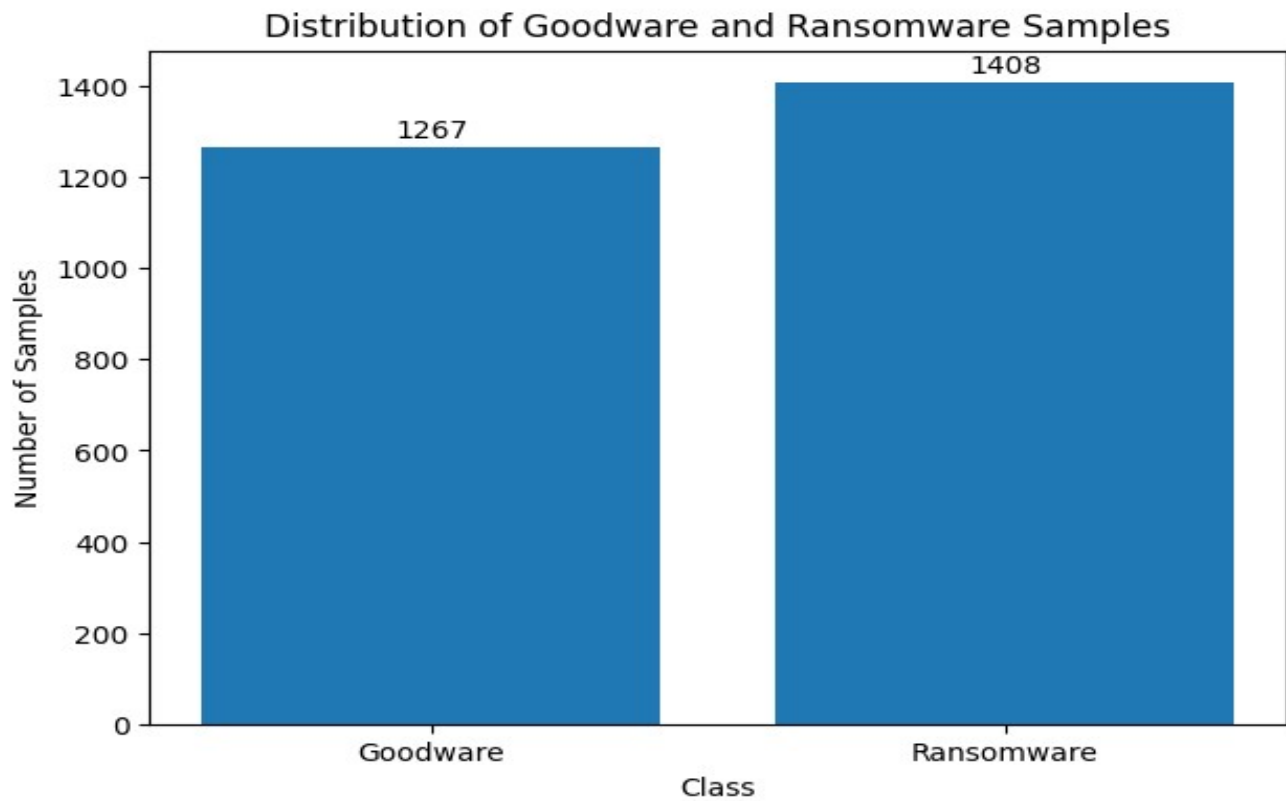
The practical execution was finalised in Python, utilising Pandas and NumPy for data manipulation, Scikit-learn for the Random Forest baseline and assessment metrics, TensorFlow/Keras for the neural networks, and Matplotlib for visualisation. The concluding project adhered to the multi-metric assessment methodology suggested throughout the IPR, whereby Accuracy, Precision, Recall, F1-Score, ROC-AUC, and false-positive behaviour were recognised as essential indicators of cybersecurity efficacy.

5.2 Dataset and Exploratory Analysis Results

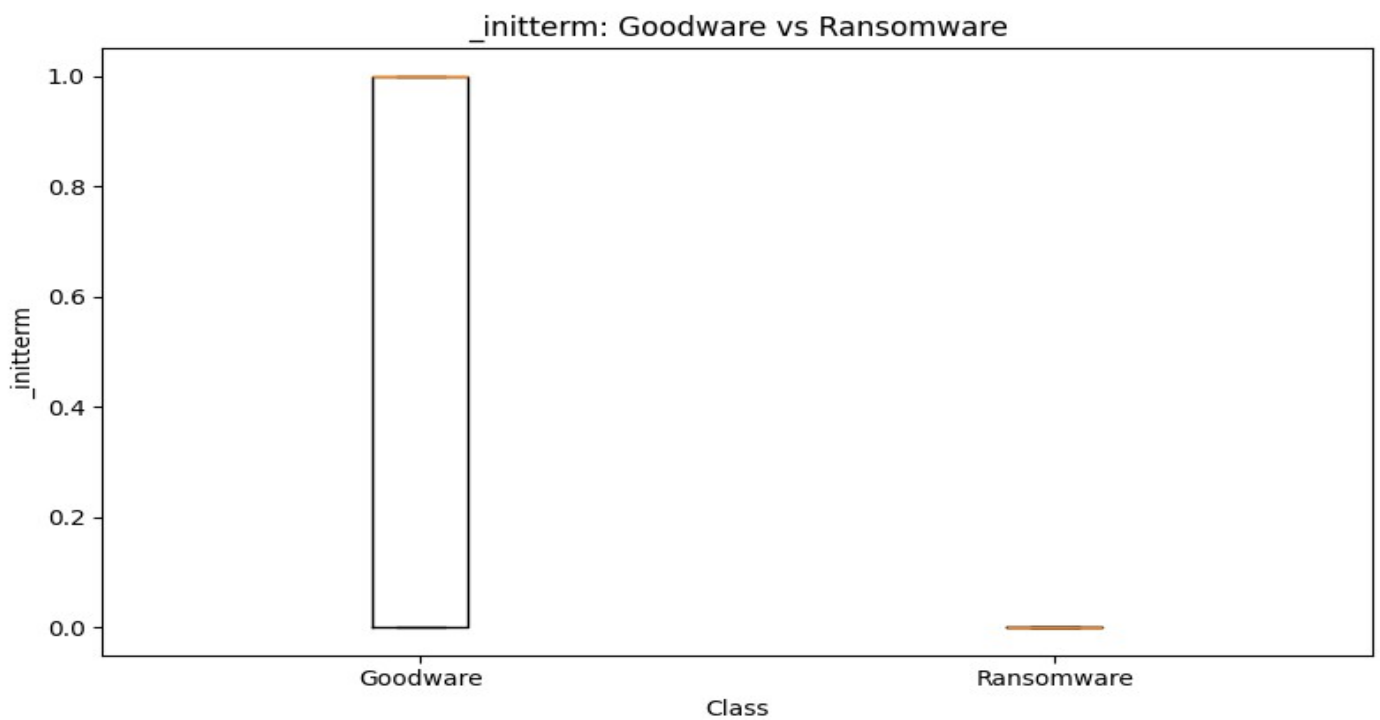
Exploratory Data Analysis revealed that the binary categories were nearly equivalent. Class imbalance may obscure inadequate detection when solely relying on accuracy. Recent investigations into ransomware emphasise that the attributes of datasets and the representation of features significantly influence the performance metrics of models (Urooj et al., 2022; Kaur et al., 2024).

Numerous structural characteristics exhibited significant correlations with the ransomware target. The correlation analysis emphasised `_initterm`, `_xcpfilter`, `_exit`, `exit`, `msvcrt.dll`, `_cexit`, `__set_app_type`, `__setusermatherr`, `memset`, and associated variables. The distinct Goodware versus Ransomware graph for `_initterm` exhibited notably robust class differentiation. This outcome is beneficial as it indicates that executable structure harbours data that may aid in ransomware identification. Nevertheless, such pronounced division necessitates vigilance. A feature may possess strong prediction capabilities as it embodies a true malware attribute, although it may also indicate the methodology employed in dataset creation. Consequently, these findings were regarded as predicting correlations instead of direct causal proof.

Class	Samples	Percentage
Goodware	1,267	47.36%
Ransomware	1,408	52.64%
Total	2,675	100%



5.2 Dataset and Exploratory Analysis Results



5.2 Dataset and Exploratory Analysis Results

5.3 Final Model Performance

The most compelling proof of model efficacy was derived from the autonomous final test set of 535 samples, which included 253 Goodware and 282 Ransomware instances. Three models were evaluated: Random Forest, DNN Model 1, and DNN Model 2.

Model	TN	FP	FN	TP	Accuracy	Ransomware Recall	ROC-AUC	Average Precision
Random Forest	246	7	2	280	98.32%	99.29%	0.9997	0.9997
DNN Model 1	243	10	3	279	97.57%	98.94%	0.9947	0.9937
DNN Model 2	246	7	6	276	97.57%	97.87%	0.9952	0.9950

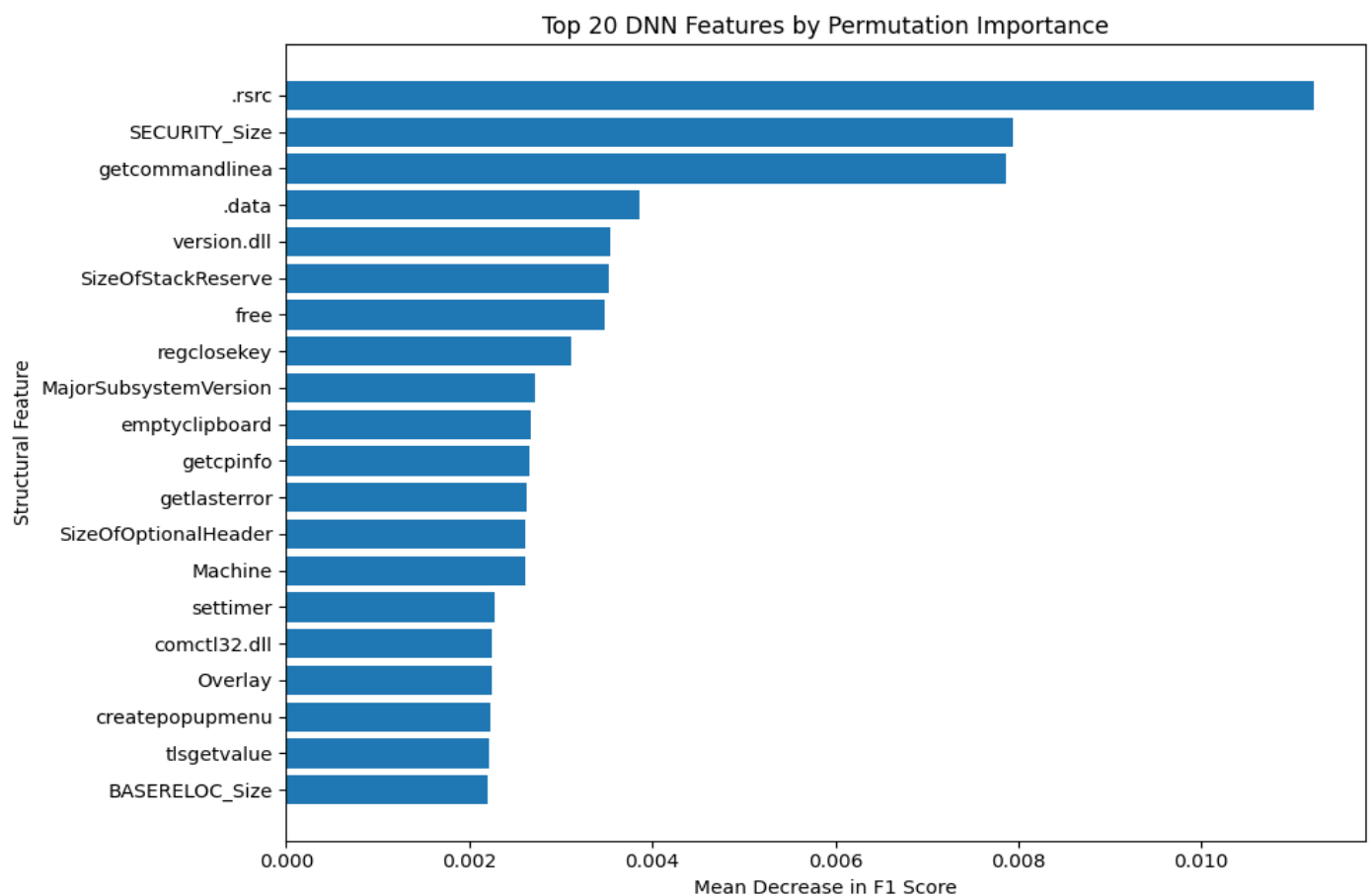


Figure 5.3: Random Forest Final Test Confusion Matrix

The Random Forest algorithm yielded the most robust overall outcome. It accurately categorised 526 out of 535 test specimens, overlooking merely two ransomware instances. DNN Model 1 overlooked three ransomware instances but generated ten erroneous alarms, whereas DNN Model 2 diminished false alerts to seven while elevating false negatives to six. These findings indicate that deep learning proved to be quite effective, although it did not surpass the simpler Random Forest benchmark. This is a very essential discovery as previous studies indicate exceptionally high accuracy in deep-learning ransomware, although emphasise

that performance is largely contingent on dataset architecture and experimental methodology (Kaur et al., 2024).

5.4 ROC and Precision-Recall Results

The ROC analysis validated that every model distinguished the two categories exceptionally effectively. The Random Forest attained an AUC of 0.9997, while DNN Model 1 reached 0.9947, and DNN Model 2 recorded 0.9952. In a similar vein, Average Precision was recorded as 0.9997 for Random Forest, 0.9937 for DNN1, and 0.9950 for DNN2.

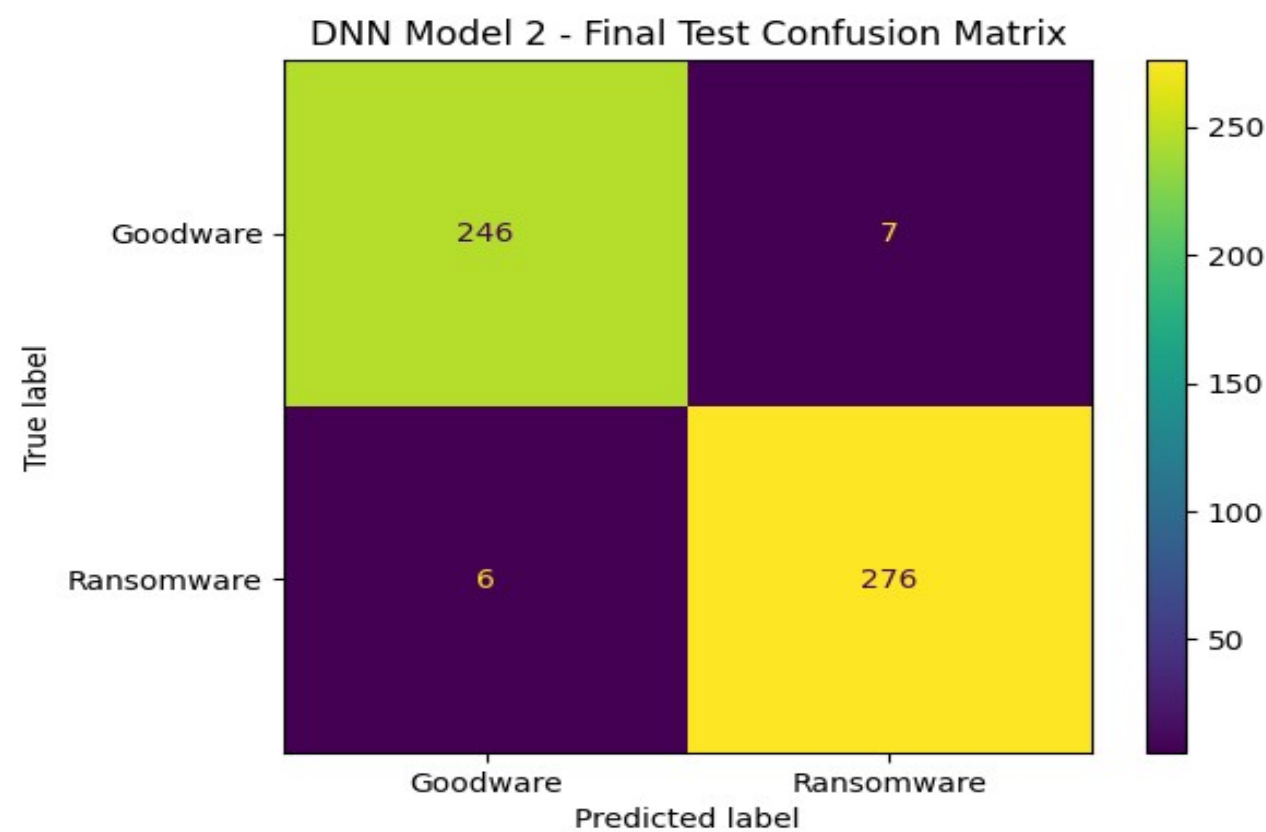


Figure 5.4: ROC Curve Comparison of Detection Models

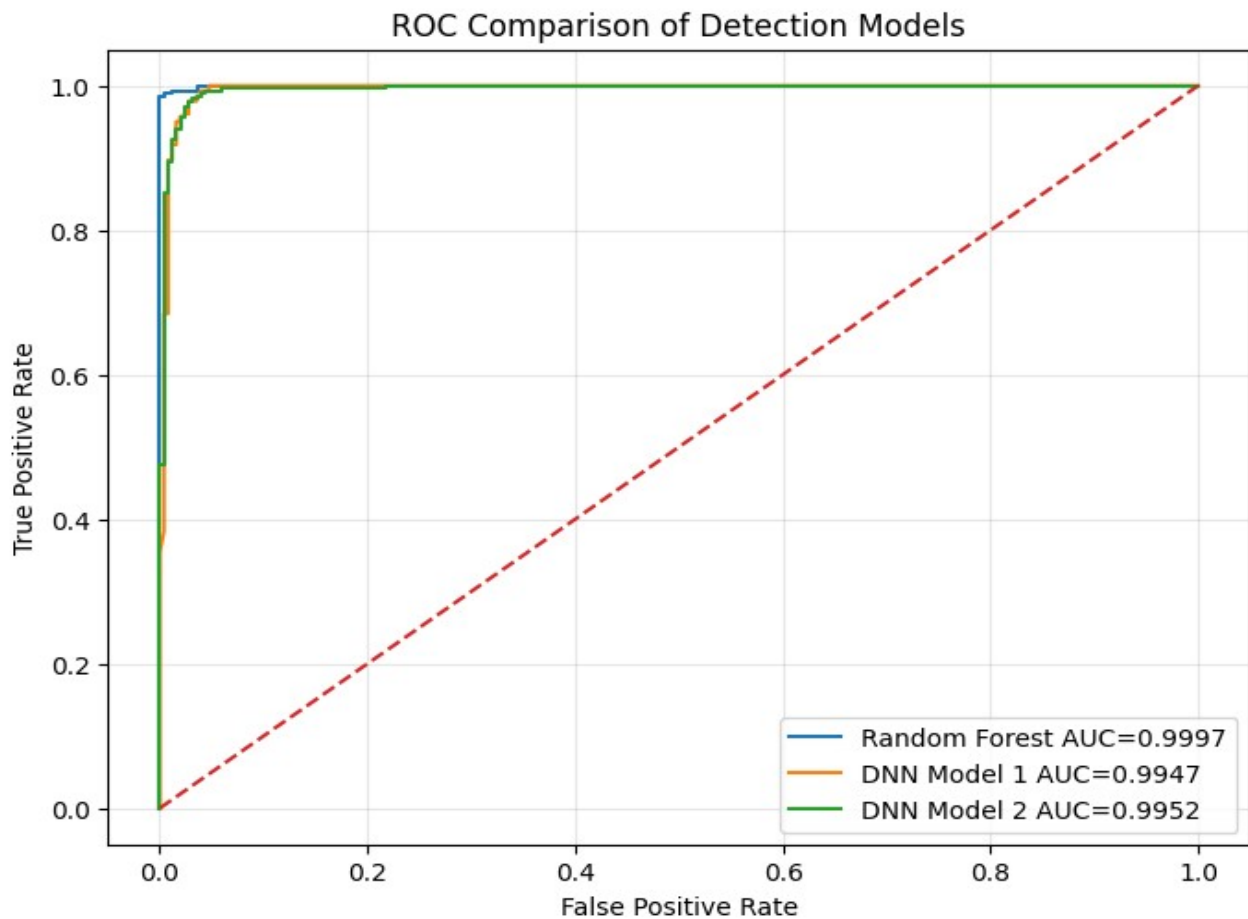


Figure 5.5: Precision–Recall Comparison of Detection Models

The disparity among the models is quantitatively minimal, nevertheless it has significance for the study inquiry. A complex model was unnecessary to attain optimal differentiation on this organised sample. Carrier et al. (2022) likewise documented exceptionally high malware-detection efficacy derived from memory and structural attributes utilising traditional machine-learning methodologies. The current findings thus endorse the perspective that the intricacy of a model must be substantiated by data instead than being presumed to enhance detection.

5.5 Threshold Analysis and Operational Trade-Off

A significant empirical study investigated the impact of the DNN decision threshold on ransomware recall, as well as the rates of false positives and false negatives. At diminished thresholds, the model preserved exceptionally high ransomware recall while generating an increased number of false alarms. As the threshold rose, the incidence of false positives diminished, while the count of overlooked ransomware instances escalated. Throughout the evaluated spectrum, false positives decreased from around eight to three, whereas false negatives rose from about one to five at elevated thresholds.

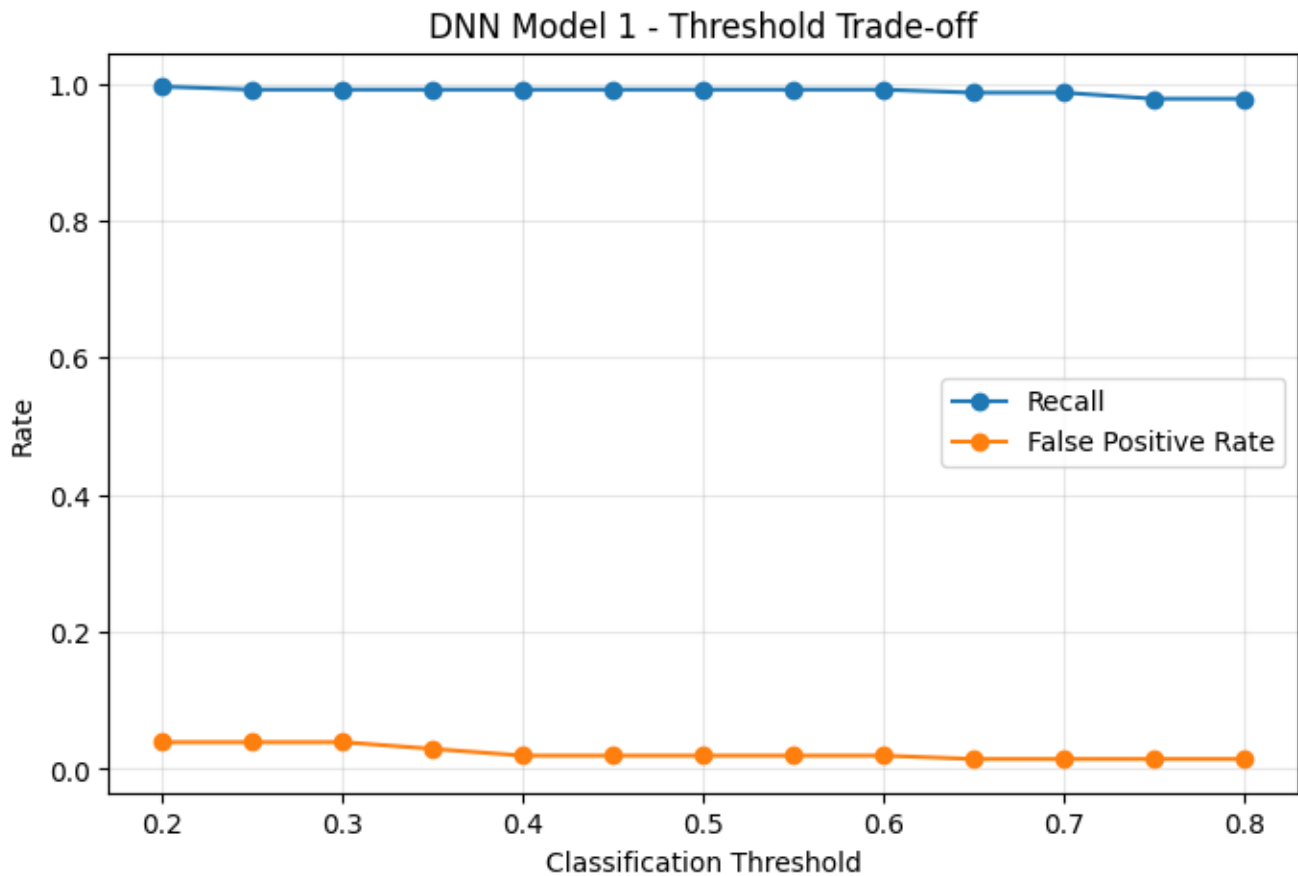


Figure 5.6: False Alerts and Missed Ransomware Across Classification Thresholds

This outcome is significant for SOC application as the optimal mathematical threshold could not correspond to the most effective operational threshold. A false positive wastes the analyst's work, but a false negative could permit ransomware to persist without scrutiny. NIST (2024) underscores the need of risk-based prioritisation in cybersecurity activities; hence, the selection of thresholds must align with the organization's capacity for overlooking risks and managing alert frequency. The initiative enhances worth by scrutinising this trade-off rather than merely adopting the standard 0.50 benchmark without assessment.

5.6 Ransomware-Family and Confidence Analysis

Family-level assessment was employed to ascertain if superior overall performance masked diminished identification of specific ransomware factions. The majority of ransomware groups attained a recall of 1.00, indicating that all test samples from these groups were identified. Family IDs 28, 23, and 15 attained recall rates of roughly 0.75, 0.83, and 0.89, respectively.

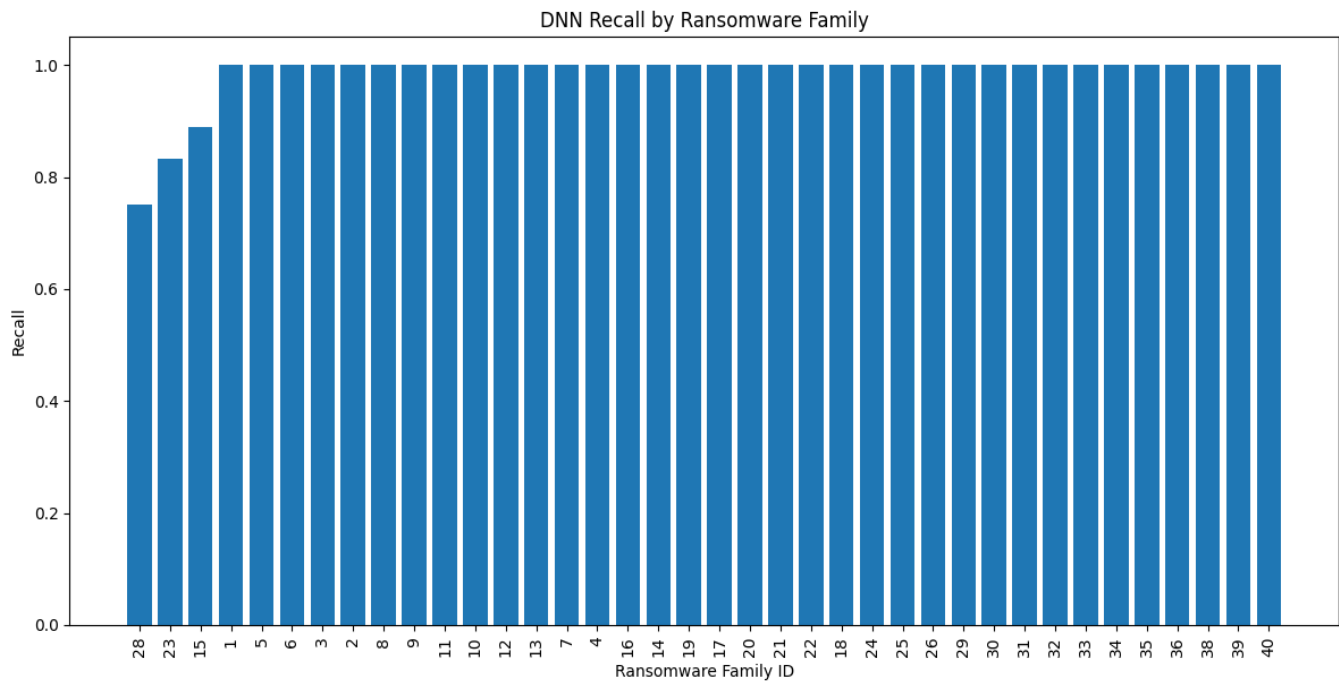


Figure 5.6: Recall by Ransomware Family

This outcome represents a significant constraint identified during the evaluation. A model exhibiting over 97% overall accuracy may yet have much poorer performance for a smaller or less representative subset. Urooj et al. (2022) recognise the evolving nature of ransomware demographics and restricted generalisability as persistent research obstacles. The findings at the family level consequently offer more compelling evidence of model quality than a singular accuracy metric.

The assessment of prediction confidence yielded a pertinent discovery. Accurate forecasts were predominantly clustered near a confidence level of 1.0, whereas erroneous predictions were roughly spread between 0.71 and 1.00, with a median nearing 0.96. Consequently, the degree of uncertainty was elevated among the errors, while certain erroneous forecasts maintained a high level of confidence.

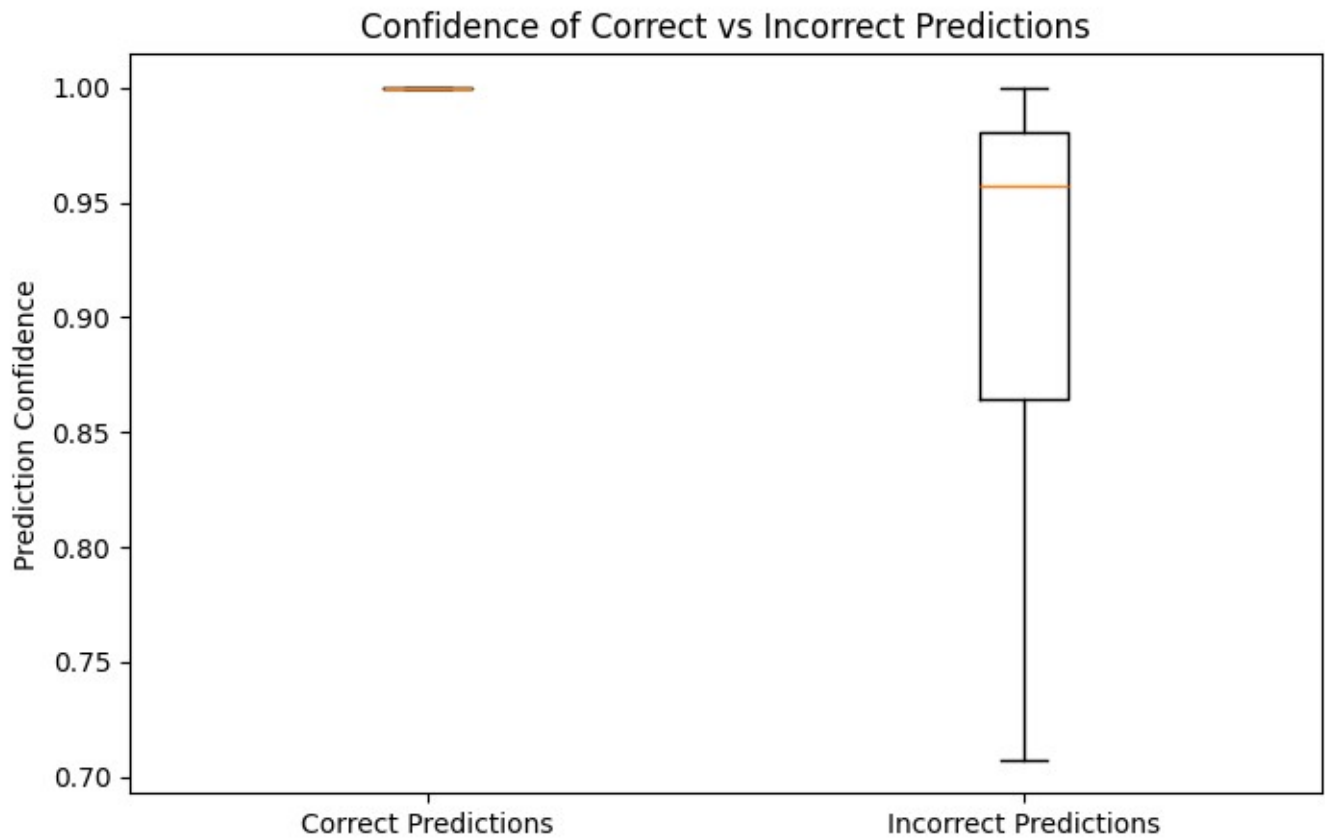


Figure 5.8: Prediction Confidence for Correct and Incorrect Classifications

This indicates that confidence can aid in SOC prioritising, although it should not be regarded as absolute certainty. NIST (2023) emphasises that credible AI necessitates accuracy, dependability, and suitable understanding of constraints. The experiment thus endorses the necessity of human analyst supervision instead of relying solely on automated responses derived from a probability score.

5.7 Feature Importance and SOC Risk Results

Permutation importance was used to examine which variables most influenced DNN predictive performance. .rsrc produced the largest reduction in F1-score when permuted, followed by SECURITY_Size and getcommandlinea. Other influential features included .data, version.dll, SizeOfStackReserve, regclosekey, Machine and SizeOfOptionalHeader.

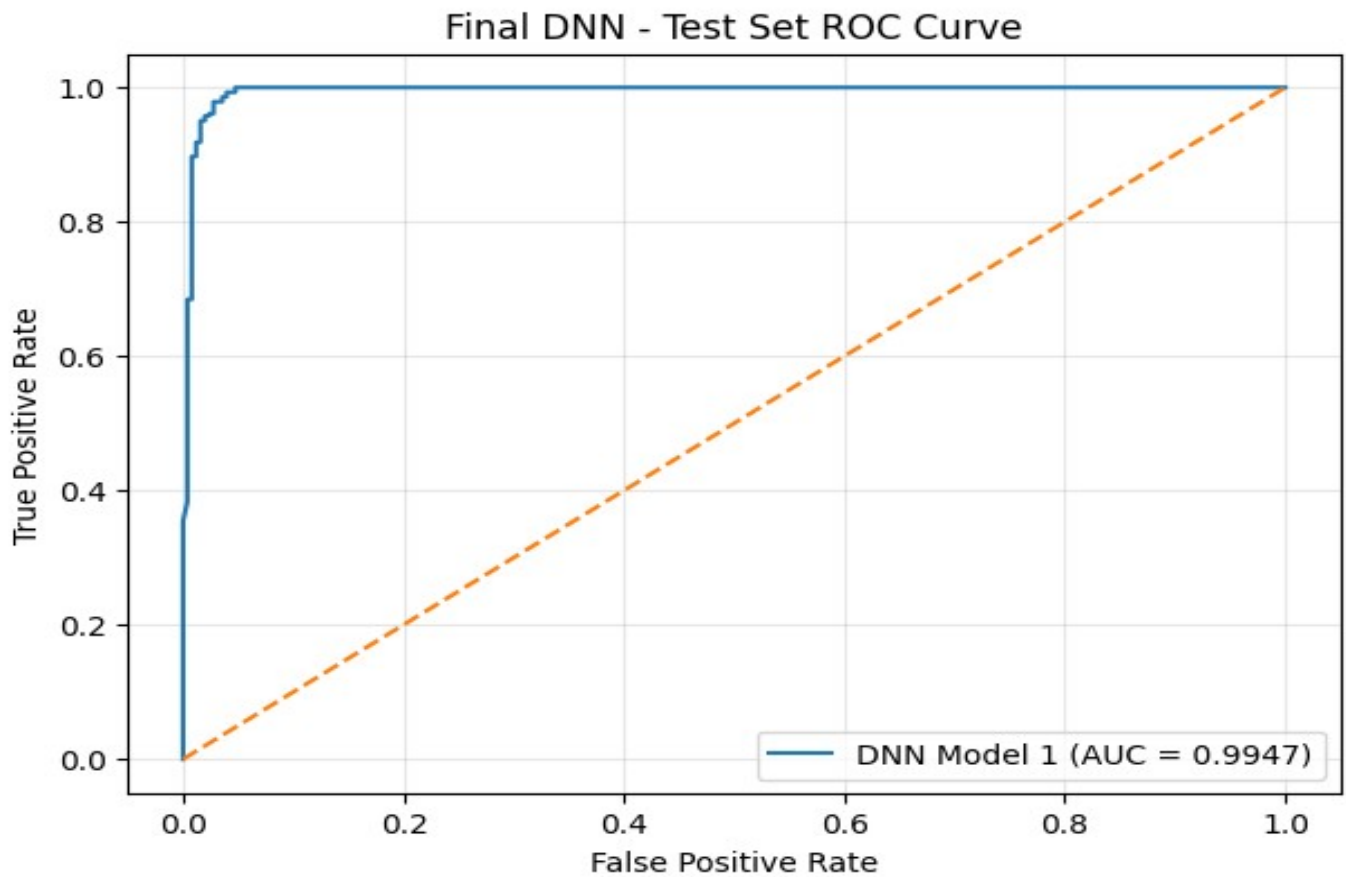


Figure 5.9: Top 20 DNN Features by Permutation Importance

These findings introduce a component of elucidation to the initiative. They indicate the features upon which the model relies, although they do not establish that these features are the source of ransomware. The distinction between correlation rankings and permutation rankings is significant: correlation indicates a direct statistical relationship, while permutation importance assesses the impact on the performance of a trained model.

The SOC risk element transformed model outputs into categories of Low, Medium, High, and Critical. The illustrated outcomes indicated a significant concentration in the Low and Critical categories, with comparatively few occurrences in the Medium and High groupings.

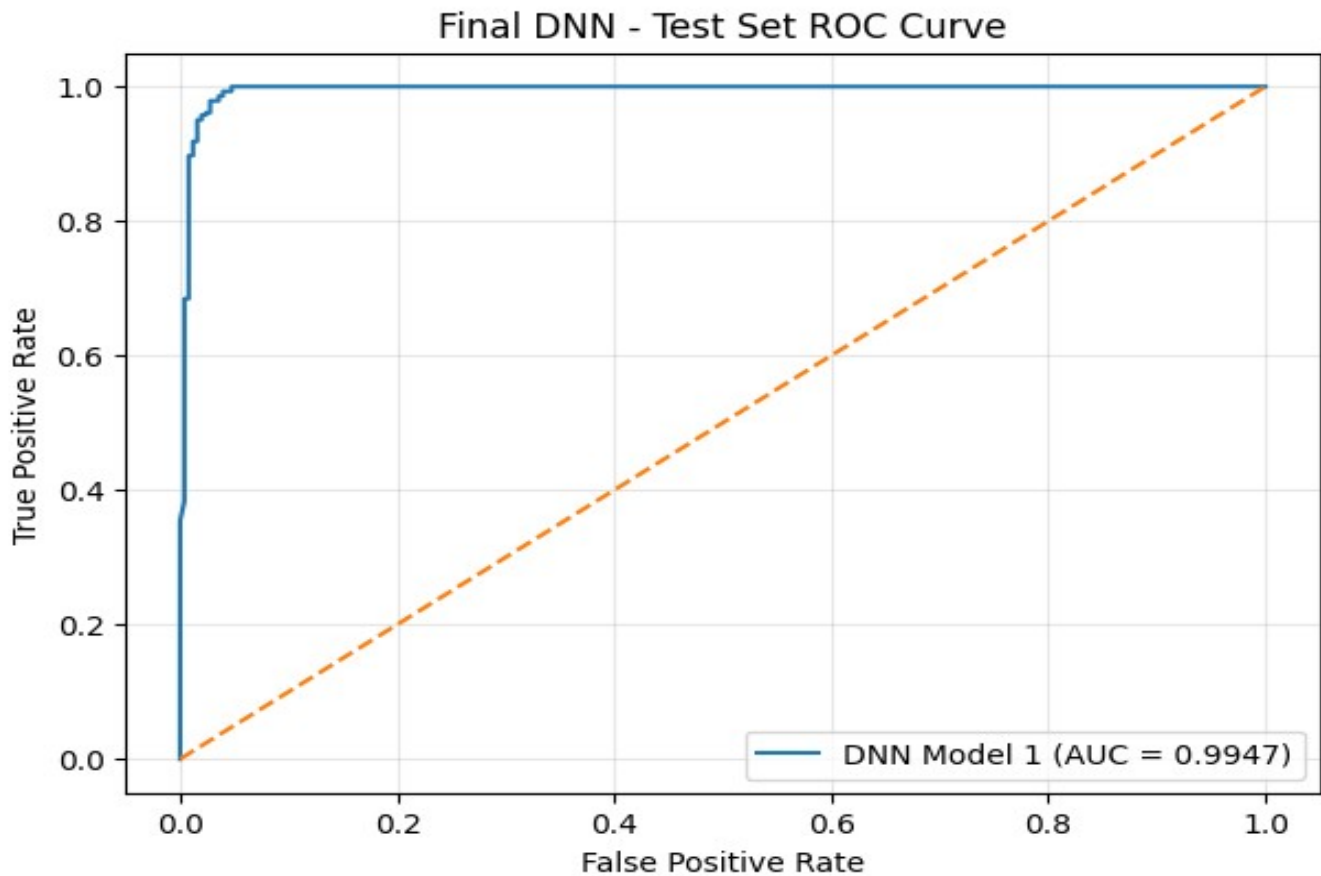


Figure 5.10: SOC Risk-Level Distribution of Test Samples

The risk layer illustrates effective incorporation beyond just binary classification, which was an initial goal of the project. Nonetheless, the uneven distribution indicates that the current thresholds might necessitate adjustment prior to actual implementation.

5.8 Overall Quality, Novelty and Feasibility

The item successfully attained robust technical outcomes and fulfilled the primary detection and assessment goals. The innovation is not primarily in developing a new neural network architecture but rather in integrating baseline comparisons, two DNN experiments, independent testing, threshold sensitivity, family-level robustness, confidence analysis, feature interpretation, and SOC risk categorisation into a single workflow.

The primary technical obstacle was steering clear of a simplistic understanding of exceptionally high model scores. Subsequent evaluations revealed that outstanding aggregate measurements may obscure familial deficiencies and significantly assured inaccuracies. This essential evaluation bolstered the research by altering the conclusion from “the models are nearly flawless” to the more justifiable assertion that the models exhibit exceptional performance on the chosen dataset nonetheless necessitate external validation.

The instruments and techniques were viable within the parameters of the project. Python and its open-source libraries offered adequate functionality without the need for costly commercial infrastructure. The findings

indicate that Random Forest could provide a pragmatic benefit where computational efficiency and robust tabular performance are crucial. The DNN models retain their significance due to their demonstrated sensitivity to ransomware and their facilitation of more in-depth tests with confidence and threshold behaviour. Collectively, the findings affirmatively address the research inquiry, indicating that practical applicability relies on factors beyond mere headline precision.

6. Evaluation and Conclusion

6.1 Final Evaluation of the Project

The initiative effectively created and assessed an AI-enhanced ransomware detection system that included machine learning, deep learning, and SOC-focused risk evaluation. The primary research inquiry examined the efficacy of deep learning in detecting ransomware-associated harmful activities and the potential of its predictions to aid in incident prioritisation. The experimental results yield a favourable yet conditional response. Deep learning demonstrated exceptional detection efficacy; nevertheless, the trials indicated that increased model complexity did not necessarily yield optimal outcomes.

The concluding test set had 535 samples, consisting of 253 Goodware and 282 Ransomware instances. Random Forest attained the highest overall efficacy with an accuracy of 98.32%, a ransomware recall of 99.29%, a ROC-AUC of 0.9997, and an Average Precision of 0.9997. Merely two ransomware specimens were overlooked. DNN Model 1 attained an accuracy of 97.57%, a ransomware recall of 98.94%, and a ROC-AUC of 0.9947, whereas DNN Model 2 reached an accuracy of 97.57%, a recall of 97.87%, and a ROC-AUC of 0.9952. These metrics indicate that all three approaches were exceptionally efficient on the chosen dataset.

Model	Accuracy	Ransomware Recall	False Positives	False Negatives	ROC-AUC	Average Precision
Random Forest	98.32%	99.29%	7	2	0.9997	0.9997
DNN Model 1	97.57%	98.94%	10	3	0.9947	0.9937
DNN Model 2	97.57%	97.87%	7	6	0.9952	0.9950

The findings thus do not endorse a straightforward assertion that deep learning need to supplant traditional machine learning. They illustrate the significance of aligning the methodology with the data. Recent studies on ransomware indicate that deep-learning methods exhibit robust efficacy; nevertheless, they also acknowledge that factors such as architectural complexity, dataset integrity, feature representation, and experimental design significantly affect results (Kaur et al., 2024). The current findings confirm this perspective as Random Forest exhibited marginally superior performance on the structured features employed in this study.

6.2 Evaluation Against the Project Objectives

The majority of the initial project goals were accomplished via practical execution and evaluation. The initiative advanced past a fundamental binary classifier by incorporating threshold analysis, ransomware-family evaluation, confidence assessment, feature interpretation, and SOC-focused risk evaluation.

Project Objective	Final Outcome
Review recent ransomware and AI-based detection research	Achieved
Prepare and analyse the ransomware dataset	Achieved
Conduct preprocessing and EDA	Achieved
Develop a conventional ML baseline	Achieved
Develop and evaluate DNN models	Achieved
Compare models using multiple metrics	Achieved
Investigate classification thresholds	Achieved
Evaluate ransomware-family robustness	Achieved
Perform feature-importance analysis	Achieved
Apply SOC-oriented risk categorisation	Achieved
Critically assess reliability and limitations	Achieved

A significant enhancement from the initial project planning was the choice to refrain from utilising a method just due to its technical intricacy. Prior planning evaluated multiple deep-learning frameworks, encompassing LSTM. Nevertheless, the ultimate dataset was predominantly organised rather than linear. Consequently, the initiative focused on Deep Neural Networks and Random Forests, which were more suitably aligned with the data. This choice enhanced methodological rigour and showcased essential model selection instead of superfluous intricacy.

6.3 Project Management and Delivery

The initiative employed a blended management approach that integrated scheduled milestones with progressive technical advancement. This methodology was beneficial as the primary submission deadlines necessitated organised advancement, whereas machine-learning development need iterative testing. The examination of the dataset, preprocessing, model formulation, and assessment cannot be regarded as entirely autonomous tasks. Exploratory results influenced subsequent feature analysis, whereas model outputs necessitated more threshold and confidence experiments.

A primary problem in management was regulating the project scope. The original idea encompassed several deep-learning methodologies, risk assessment, SOC alert creation, and prototype formulation. Endeavouring to use all conceivable algorithms would have diminished the thoroughness of assessment. The focus was consequently narrowed to a limited selection of validated models and more rigorous testing. This compromise was advantageous as the completed project encompassed three model comparisons, independent testing, ROC and precision-recall analysis, threshold evaluation, family-level recall, and feature interpretation.

A subsequent managerial insight was the significance of recording and documenting in conjunction with execution. Experimental results, graphs, and statistical metrics were documented throughout the development process rather than being recreated at the conclusion. This enhanced traceability connects the code, outcomes, and parts of the final report. It additionally endorsed the iterative methodology advocated for data science initiatives, wherein outcomes may necessitate the reevaluation of prior assumptions and design decisions (Martínez-Plumed et al., 2021).

In general, the execution of the project was feasible given the accessible scholarly resources. The investigation was accomplished without the use of commercial modelling tools, thanks to Python, Scikit-learn, and TensorFlow/Keras. The initiative therefore shown that significant ransomware investigation may be conducted using publicly accessible data and open-source resources.

6.4 Key Technical and Research Insights

A multitude of significant revelations were acquired from the inquiry. The initial point is that improved precision does not convey the entire narrative of cybersecurity. All three models attained an accuracy ranging from 97.6% to 98.3%, however the confusion matrices indicated varying operational characteristics. DNN Model 1 generated ten false positives and three false negatives, while DNN Model 2 decreased the false positives to seven but increased the false negatives to six. The Random Forest algorithm generated seven false positives and merely two false negatives.

This discovery gained prominence during threshold evaluation. Elevating the categorisation threshold diminished superfluous alarms, while concurrently heightened the incidence of overlooked ransomware samples. The experiment therefore illustrated a tangible compromise between security sensitivity and SOC effort. NIST (2024) underscores that cybersecurity choices ought to be founded on risk assessment rather than only on technical solutions. The threshold experiment corroborates this idea as there exists no universally applicable probability threshold for all organisations.

The second significant revelation emerged from the research of malware families. Despite the overall DNN recall being exceptionally high, family IDs 28, 23, and 15 attained recalls of roughly 0.75, 0.83, and 0.89, respectively, whereas the majority of other families earned a perfect score of 1.00. This verifies that overall performance may obscure specific deficiencies. A production detector must be evaluated not just against a unified harmful category but also against emerging families and evolving malware distributions.

The third revelation pertained to the assurance of the model. Accurate classifications typically exhibited elevated confidence levels; however, erroneous predictions also displayed confidence values nearing 1.0, with the median confidence for these inaccuracies being 0.96. Thus, probability should not be regarded as certainty. This discovery reinforces the premise of the NIST AI Risk Management Framework that emphasises the necessity of human supervision, validation testing, and clear disclosure of uncertainty in AI systems (NIST, 2023).

6.5 Comparison with Existing Literature

The findings largely align with contemporary studies indicating that machine-learning and deep-learning methodologies can attain exceptional ransomware detection efficacy. Urooj et al. (2022) recognised that integrating behavioural and dynamic analysis with machine learning offers a viable solution to the

shortcomings of signature-based detection. Kaur et al. (2024) furthermore discovered that contemporary research on deep-learning ransomware frequently indicated elevated detection rates, with some instances nearing 99%.

Nonetheless, the current initiative introduces a significant analytical viewpoint. Published findings may foster the presumption that deep learning is a natural advancement over traditional machine learning. The present study did not detect this result. The Random Forest model attained a ROC-AUC of 0.9997, surpassing both DNN setups. Carrier et al. (2022) further illustrated that meticulously designed malware-associated attributes might facilitate exceptional classification efficacy using traditional machine learning. Consequently, the current research substantiates the claim that the appropriateness of models have to be evaluated by empirical methods.

The initiative additionally endorses material that cautions against generalisation. Urooj et al. (2022) recognise alterations in the ransomware population as a critical issue, since novel variants may exhibit distinct behaviours compared to the samples utilised in training. The diminished recollection seen for several malware families in this study offers concrete experimental proof of this problem.

The results of confidence are also connected to the broader issue of the opaque characteristics of deep learning. Kaur et al. (2024) recognise interpretability as a significant constraint in ransomware detection. Permutation significance partially tackled this by recognising `.rsrc`, `SECURITY_Size`, and `getcommandlinea` as significant attributes. Nonetheless, feature significance does not establish that these characteristics serve as universal markers of ransomware. The findings must consequently be understood within the confines of the chosen dataset.

6.6 Challenges, Limitations and Feasibility

The primary constraint is external validity. The models underwent training and evaluation on a singular curated dataset. Superb results on the reserved test data may not ensure comparable outcomes against forthcoming ransomware, varying operating conditions, or independently gathered datasets. Cross-dataset validation would consequently yield more compelling evidence than more optimisation on the current dataset.

A secondary constraint pertains to exceptionally robust feature correlations. Variables like `_initterm` and `_xcpfilter` had significant connections with the ransomware objective. These may signify valuable executable traits, although exceptionally robust prediction patterns can also emerge from the assembly of the dataset. Subsequent research ought to perform a systematic evaluation of leakage and dataset bias prior to asserting more robust conclusions regarding real-world generalisation.

The risk analysis of the SOC represents an additional constraint. The methodology effectively categorised model outputs into Low, Medium, High, and Critical classifications, although the results were predominantly focused on the Low and Critical extremes. This implies that the existing risk weights and thresholds can be excessively polarised. In a functional SOC, risk must take into account asset worth, business repercussions, detected attack patterns, vulnerability data, and threat intelligence. Consequently, the existing risk framework illustrates practicality rather than an authenticated commercial risk apparatus.

Notwithstanding these constraints, the methodology proved viable for a Master's research endeavour. The dataset comprising 2,675 observations was feasible to handle with the existing computational resources, allowing for model training and evaluation without the need for specialised hardware or costly commercial platforms. This renders the artefact authentic as a research prototype; but, practical implementation would necessitate even more robust validation, security engineering, and integration.

6.7 Commercial and Economic Evaluation

The identification of ransomware holds significant business importance, as effective assaults can lead to operational interruptions, expenses for recovery, regulatory repercussions, and damage to reputation. A security framework delivers economic benefits solely when it mitigates risk without imposing undue operational burdens. This becomes the trade-off between false positives and false negatives commercially significant.

DNN Model 1 identified 279 out of 282 ransomware specimens, although produced 10 erroneous alarms. A minor rise in false positives can be tolerable for an organization where ransomware interruptions would incur significant costs. In contrast, a high-throughput SOC might prioritise the Random Forest outcome due to the generation of merely seven false positives while failing to detect only two ransomware instances.

The open-source implementation additionally enhances practicality. Python, Scikit-learn, and TensorFlow diminish software license expenses and facilitate the reproduction or enhancement of the methodology. Nonetheless, a production system would necessitate extra costs for oversight infrastructure, secure implementation, model upkeep, threat intelligence incorporation, and analyst education. Consequently, one should not deduce economic worth solely from laboratory precision.

6.8 Recommendations and Future Work

The foremost priority for next endeavours is autonomous verification. The models ought to be assessed with malware samples from various origins and timeframes to ascertain if their performance remains consistent during population drift.

Subsequent trials ought to incorporate probability calibration, as the confidence assessment revealed that erroneous predictions can still exhibit considerable confidence. Supplementary interpretability techniques, such as SHAP, may offer localised elucidations for specific warnings and enhance the overarching permutation-importance assessment.

The testing of ransomware families had to be broadened, especially for those that demonstrated diminished recollection. Data augmentation or focused training may ascertain if these vulnerabilities stem from limited sample numbers or authentically distinct structural traits. Monitoring for concept drift would be advantageous due to the ever-evolving nature of ransomware.

The SOC risk framework ought to be broadened beyond mere model reliability. Subsequent iterations may integrate detection likelihood with asset significance, attack-phase data, threat intelligence, and organisational repercussions. The resultant risk assessment ought to be evaluated by SOC professionals to ascertain if the priority rankings are significant and practical.

Ultimately, research focused on deployment should investigate inference duration, memory utilisation, real-time file processing, and model resilience to adversarial alterations. These evaluations would transition the initiative from a robust academic prototype to a functional cybersecurity framework.

6.9 Final Conclusion

This initiative examined the application of artificial intelligence for identifying ransomware and the incorporation of predicting outcomes with risk prioritisation tailored for Security Operations Centers. The finalised artefact effectively integrated data preparation, exploratory analysis, Random Forest classification, two DNN configurations, multi-metric evaluation, threshold testing, ransomware-family analysis, prediction-confidence analysis, feature significance, and SOC risk categorisation.

The primary study inquiry may be affirmatively addressed: AI-driven models demonstrated significant efficacy in differentiating Goodware from Ransomware within the chosen dataset, and their results could be converted into actionable insights for SOC-style prioritisation. Nonetheless, the results indicate that deep learning was not inherently the most effective method. The Random Forest model attained the highest overall test performance, achieving an accuracy of 98.32%, a ransomware recall of 99.29%, a ROC-AUC of 0.9997, and merely two false negatives.

The project's significant impact extends beyond merely achieving a high categorisation score. Threshold investigation indicated that diminishing false warnings may lead to an increase in overlooked ransomware. Testing at the family level uncovered deficiencies obscured by collective metrics. Confidence assessments revealed that erroneous forecasts may nonetheless exhibit a strong sense of certainty. Feature analysis enhanced comprehension yet simultaneously provoked enquiries regarding dataset reliance.

The project successfully achieved its main goals and showcased a technically viable AI-assisted ransomware detection system. Simultaneously, the findings endorse a prudent strategy for operational implementation. The system ought to be viewed as a decision aid for security analysts instead of a self-sufficient substitute for human discernment. Additional external validation, calibration of risk scores, and empirical testing in real-world scenarios would be necessary prior to commercial implementation. In the context of an MSc research endeavour, the study offers a significant amalgamation of ransomware identification, essential model assessment, and pragmatic SOC-focused risk evaluation.

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8. Appendices

Appendix A – Dataset Summary

Table A.1: Dataset Overview

Item	Details
Project task	Binary ransomware detection
Target classes	Goodware and Ransomware

Total samples	2,675
Goodware samples	1,267
Ransomware samples	1,408
Goodware percentage	47.36%
Ransomware percentage	52.64%
Data type	Structured malware/software features
Main feature categories	Executable structure, PE-related characteristics, imported functions/libraries and related software characteristics
Main use	Training and evaluating ransomware detection models
Additional information	Ransomware-family information used for family-level robustness testing

Table A.2: Class Distribution

Class	Samples	Percentage
Goodware	1,267	47.36%
Ransomware	1,408	52.64%
Total	2,675	100%

Appendix B – Data Preparation and Exploratory Analysis

Table B.1: Data Preparation Process

Stage	Activity	Purpose
Data loading	Dataset imported into Python	Prepare data for analysis
Structural inspection	Dimensions, columns and data types examined	Understand dataset structure

Missing-value inspection	Missing values checked	Identify incomplete observations
Duplicate inspection	Duplicate records examined	Reduce repeated information
Target preparation	Goodware/Ransomware labels prepared	Enable supervised learning
Feature inspection	Numerical and structural variables examined	Identify useful predictors
Class analysis	Target distribution calculated	Assess class balance
Correlation analysis	Relationships between features and target examined	Identify strong associations
Feature preparation	Predictors separated from target	Prepare modelling inputs
Scaling	Applied where required for DNN modelling	Improve neural-network optimisation
Data splitting	Development and independent testing data separated	Support reliable evaluation

Table B.2: Selected Features Identified During Exploratory Analysis

Feature	Observation	Relevance
_initterm	Strong relationship with target	Potential structural discriminator
_xcpfilter	Strong relationship with target	Useful classification information
_exit	Associated with target class	Potential malware-related structural pattern
exit	Class-related variation observed	Supports structural analysis
msvcrt.dll	Association observed	Library/import information
_cexit	Target relationship observed	Potential predictive information
__set_app_type	Structural relationship identified	Candidate predictive feature
__setusermatherr	Association with target	Candidate predictive feature
memset	Class-related pattern identified	Supporting predictor
.rsrc	High permutation importance	Strong contribution to model performance

Appendix C – Model Development

Table C.1: Models Implemented

Model	Type	Role in Project	Main Reason for Selection
Random Forest	Machine learning ensemble	Baseline classifier	Strong performance on structured tabular data
DNN Model 1	Deep Neural Network	Main DL experiment	Evaluate non-linear feature relationships
DNN Model 2	Deep Neural Network	Alternative DL configuration	Test whether architectural changes improve performance

Table C.2: Model Development Workflow

Step	Implementation
1	Load processed ransomware dataset
2	Separate input features and target
3	Create development/test partitions
4	Prepare/scaled features where appropriate
5	Train Random Forest baseline
6	Train DNN Model 1
7	Monitor training and validation behaviour
8	Train DNN Model 2
9	Generate class probabilities
10	Convert probabilities to classifications
11	Calculate evaluation metrics
12	Compare model performance
13	Perform threshold testing
14	Conduct ransomware-family analysis
15	Analyse prediction confidence
16	Perform permutation importance
17	Generate SOC-oriented risk categories

Appendix D – Final Model Performance

Table D.1: Final Independent Test Results

Model	TN	FP	FN	TP	Accuracy	Ransomware Recall	ROC-AUC	Average Precision
Random Forest	246	7	2	280	98.32%	99.29%	0.9997	0.9997
DNN Model 1	243	10	3	279	97.57%	98.94%	0.9947	0.9937
DNN Model 2	246	7	6	276	97.57%	97.87%	0.9952	0.9950

Appendix E - SOC-Based Risk Analysis

Table E.1: SOC Risk Framework

Risk Level	General Meaning	Suggested SOC Response
Low	Low indication of ransomware	Record or routine monitoring
Medium	Some suspicious evidence	Review if supporting indicators exist
High	Strong ransomware indication	Prioritised analyst investigation
Critical	Very strong ransomware indication	Immediate investigation and escalation

Appendix F – Software and Development Tools

Table F.1: Tools Used

Tool/Technology	Project Use
Python	Main implementation language
Google Colab/Jupyter environment	Interactive development and experimentation
Pandas	Dataset loading, cleaning and manipulation
NumPy	Numerical processing
Scikit-learn	Random Forest, preprocessing and model evaluation
TensorFlow/Keras	DNN development and training
Matplotlib	Experimental plots and visualisation
Permutation importance	Model feature interpretation

Confusion matrix	Error analysis
ROC curve	Discrimination analysis
Precision-recall curve	Positive-class performance evaluation

Appendix G – Literature Review Summary

Table G.1: Literature Review Matrix

Study	Research Focus	Method/Approach	Main Findings	Limitation / Gap	Relevance to This Project
Alraizza and Algarni (2023)	ML-based ransomware detection	Survey of machine-learning ransomware research	ML techniques show strong potential for ransomware classification	Differences in datasets and experiments make direct comparison difficult	Supports ML baseline and multi-model comparison
Urooj et al. (2022)	Dynamic ransomware detection	Systematic review of dynamic analysis and ML	API calls, system calls and runtime behaviour are useful for detection	Population drift and lack of common benchmarks remain challenges	Supports robustness and ransomware-family testing
Cen et al. (2024)	Early ransomware detection	Survey of early-detection techniques	Early identification is important before destructive encryption occurs	Real-time and early detection remain difficult	Supports practical motivation for rapid ransomware identification
Kritika (2025)	Deep-learning ransomware detection	Comprehensive DL literature review	DL approaches frequently achieve very high classification performance	Dataset quality, computational requirements and explainability remain concerns	Supports DNN experiments and critical interpretation
Carrier et al. (2022)	Obfuscated malware detection	Memory feature engineering with ML	Reported approximately 99% classification performance	Controlled datasets may not represent every operational environment	Supports use of conventional ML and structural features
Sarker (2021)	Deep cybersecurity	Review of neural networks and DL for	DL can model complex	Requires representative	Supports DNN selection and

		cybersecurity	cybersecurity patterns	data and suitable model design	evaluation
Sarker, Furhad and Nowrozy (2021)	AI-driven cybersecurity	AI-based security-intelligence framework/review	AI can support automated security analysis and intelligent decisions	Trust, explainability and changing threats remain issues	Supports AI-driven SOC concept
Alzubaidi et al. (2021)	Deep-learning methods	Broad review of DL architectures and applications	DL provides powerful non-linear representation capability	Computational cost and model selection are important	Supports technical justification for neural networks
NIST (2023)	Trustworthy AI and AI risk	AI Risk Management Framework	AI should be valid, reliable, secure, transparent and appropriately managed	Framework requires contextual implementation	Supports cautious interpretation of high-confidence predictions
NIST (2024)	Cybersecurity risk management	Cybersecurity Framework 2.0	Cybersecurity requires governance, identification, protection, detection, response and recovery	Framework does not prescribe one technical detection model	Supports integration of detection with SOC risk management
ENISA (2024)	Current cyber-threat landscape	European threat analysis	Ransomware remains a major cybersecurity threat	Threat landscape changes continuously	Supports project significance and need for adaptive detection
IBM (2024)	Economic impact of cyber incidents	Industry breach-cost analysis	Data breaches create substantial financial and operational costs	Industry-level findings are not ransomware-model performance measures	Supports commercial and economic justification
Verizon (2025)	Data breaches and ransomware	Large-scale incident and breach analysis	Ransomware continues to contribute significantly to security breaches	Observational breach statistics do not directly evaluate ML detectors	Supports real-world motivation
Mendes	Explainable	Systematic literature	Explainability	XAI techniques	Supports

and Rios (2023)	AI in cybersecurity	review	can improve understanding and trust in cybersecurity AI	have practical and methodological limitations	feature- importance analysis
Zhang et al. (2022)	Explainable AI for cybersecurity	Review of XAI cybersecurity applications	XAI can help analysts understand automated security decisions	Explanations may not establish causality	Supports cautious interpretation of feature importance